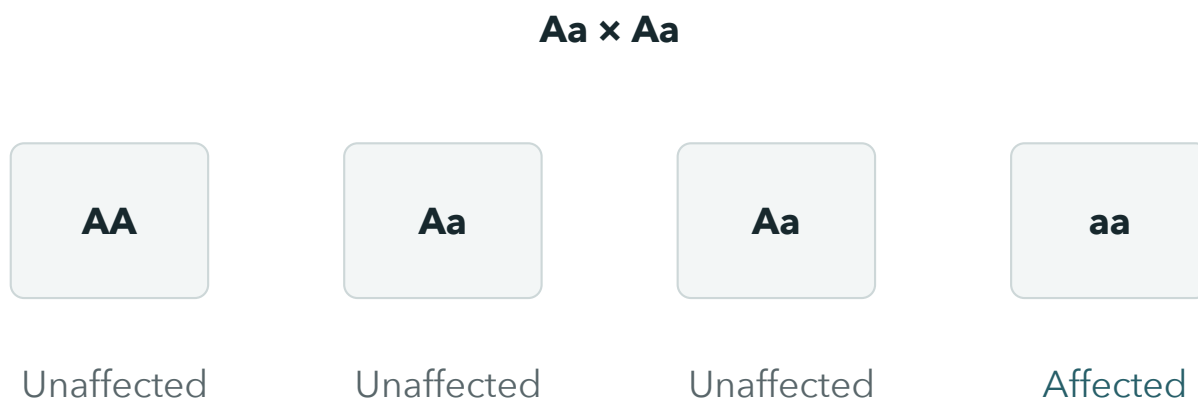


Pedigrees, penetrance and uncertain variants

Part 1 | Condition the pedigree probability correctly

Excluding an affected genotype changes the denominator of the remaining possibilities.



Among unaffected: 2 carriers / 3 outcomes

For two carriers, the unaffected offspring possibilities include two heterozygous outcomes and one unaffected homozygous outcome.

The required assumptions

For a fully penetrant autosomal-recessive model with two carrier parents, offspring probabilities are $1/4$ AA, $1/2$ Aa and $1/4$ aa. Given that an offspring is unaffected, exclude aa and renormalise:

$P(\text{carrier}|\text{unaffected}) = (1/2)/(3/4) = 2/3$. This conditioning differs from the unconditioned $1/2$ carrier probability.

Penetrance is not severity

Whether a phenotype occurs and how it appears are different properties.

Term

Feature	Meaning	Pedigree consequence
Reduced penetrance	Some variant carriers lack recognised phenotype	An unaffected carrier can transmit
Variable expressivity	Manifestations vary between affected people	Severity can differ within a family
Age-dependent recognition	Features not yet apparent	Current examination has limits

Use a model honestly

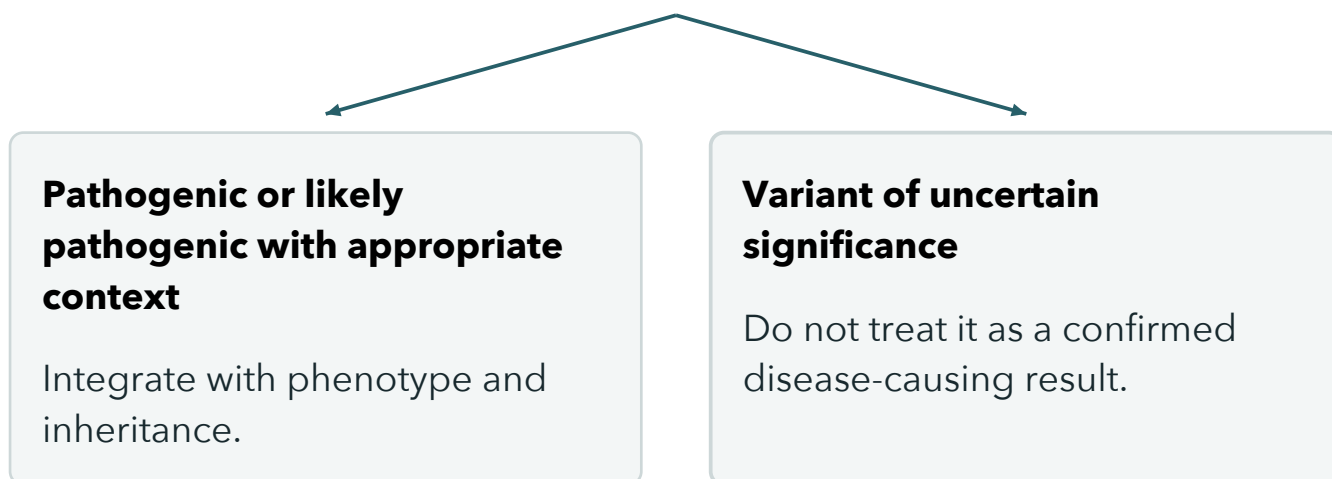
An apparently skipped generation does not automatically convert a dominant disorder into recessive inheritance. Consider penetrance, age and incomplete phenotyping before selecting a pattern, and avoid applying a fully penetrant calculation to a family explicitly described as incompletely penetrant.

A VUS is unresolved evidence

An uncertain variant is not a validated predictive result.

A variant is reported

Which category is established?



Evidence that can help

Segregation, population frequency, validated functional evidence and phenotype matching can contribute to classification. A striking clinical story does not make any rare variant causative; nor does an uncertain result rule out a genetic cause. Use the classification framework rather than rarity alone.

Genotype, transcript and protein are separate readouts

An unchanged coding sequence does not establish unchanged gene expression.

Assay

Feature	Primary level	Useful limit
DNA sequencing	Sequence variation	Not transcript abundance
RNA assay	Transcript state	Not protein function alone
Protein assay	Amount or activity	Not a complete upstream explanation

A staged investigation

When protein falls but coding sequence is unchanged, compare RNA abundance before attributing the change to translation or degradation. If RNA also falls, the experiment has narrowed the level; it has not yet identified which regulatory element or cellular process produced that change.

A negative test needs a defined scope

Testing limitations depend on what changes and tissues the method can detect.

Read a negative report

- Identify the tested genes and variant classes
↓
- Check coverage and analytical limits
↓
- Compare the phenotype and inheritance model
↓
- Decide what uncertainty remains

A useful distinction

A result can be analytically negative within a well-defined panel while leaving an untested mechanism plausible. Describe that residual uncertainty precisely. Avoid both extremes: treating every negative test as complete exclusion, or treating a high-quality targeted negative result as providing no information.